

ADITYA NARAYAN RAY

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Professional Summary

Computer Science undergraduate (9.82 CGPA) pursuing a **Software Engineer, Intern** position, with a solid foundation in data structures, algorithms, and backend system design. Track record of shipping production code: merged pull requests to a WebRTC platform used by millions, a document-processing backend handling 50,000+ records at 91% precision, and a full-stack system exposing 30+ REST endpoints. Effective at collaborating cross-functionally, incorporating code review feedback, and learning new systems quickly.

Technical Skills

Programming Languages: C++, Python, JavaScript, TypeScript, C
Backend & Systems: FastAPI, Node.js, Kafka, REST APIs
Frontend Technologies: React.js, Tailwind CSS, WebSockets, WebRTC
Databases and Storage: PostgreSQL, MySQL, MongoDB, Redis
Tools & Practices: Git, GitHub, Docker, CI/CD, System Design

Education

Manipal University Jaipur

Aug 2024 – May 2028

Bachelor of Technology in Computer Science and Engineering

Jaipur, India

- CGPA: **9.82/10.0** — **Dean's List** Recipient for **4 Semesters** — **9.67, 10, 9.72, 9.90**
- Relevant Coursework: **Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Design and Analysis of Algorithms**

Professional Experience

Summer Research Intern

May 2026 – June 2026

NISER (National Institute of Science Education and Research)

Bhubaneswar, Odisha

- Benchmarked **Weighted Nonlinear Least Squares (WNLLS)**, **Gaussian Process regression**, and **Physics-Informed Neural Networks (PINNs)** for estimating photon mass attenuation coefficients from **Cs-137 gamma-ray transmission data**; co-authoring a comparative methodology paper on the tradeoffs.
- Built and iteratively debugged PINN training pipelines across **five materials** (Al, Cu, Brass, Steel, Pb), achieving $R^2 \geq 0.99$ on Aluminum.

Open Source Contributor

October 2025 – February 2026

Jitsi - GSoC

Remote

- Contributed **merged pull requests** across **3 repositories** (jitsi-meet, lib-jitsi-meet, jitsi-meet-electron-sdk) in a **production WebRTC** platform used by millions globally.
- Developed a **PTZ (Pan-Tilt-Zoom) camera control** feature via the **WebRTC constraints API**, integrating **TypeScript** UI components and resolving a critical **uninitialized field bug** in JitsiLocalTrack; **GSoC 2026 proposal** scored **89/100** after mentor validation.

Technical Projects

Sentinel AI – NGO Rescue Coordination Platform | *React, FastAPI, PostgreSQL, Docker*

2026

- Built a full-stack platform (React, FastAPI, PostgreSQL) consolidating citizen reports, FIRs, images, and voice notes into a unified case workflow, with **JWT-based role access control** across citizen, NGO admin, police, and volunteer roles.
- Designed a modular backend exposing **30+ REST endpoints**, integrating **OCR, speech-to-text**, and **LLM-based summarization** for AI-assisted case risk prioritization, a rule-based shelter recommendation engine, and a case-tracking dashboard.

Ny – Legal Data Platform | *Python, FastAPI, PostgreSQL, RAG, Vector Search*

October 2025

- Developed an automated **document categorization pipeline** using **Python** and **FastAPI** to securely parse and map court transcripts, processing a dataset of **50,000+ records** with **91% precision**.
- Implemented a **hybrid retrieval architecture** combining **vector similarity search** and relational **PostgreSQL** queries (**RAG pipeline**), reducing system response times to under **200ms** and improving data accuracy by **35%**.

Honors, Awards & Certifications

Competition Awards: Secured **1st Place** out of **100+ competing engineering teams** at the Bug Monopoly Competition, **Evoque 2025**; Finalist at **JPMorganChase Code For Good 2026**

Academic Honors: Awarded **Dean's List** status for maintaining a **top 1% academic ranking** across **4 consecutive semesters**

Certifications: **Machine Learning Specialization** (Andrew Ng / DeepLearning.AI), **Microsoft Azure AI Fundamentals**
Leadership: **Web Development Head**, ACM MUJ SIGAI – lead and mentor a team of **40+ students**; **Buddy Mentor** for undergraduates